

PROBLEM SET 4

TRANSPORT PHENOMENA I CHE 311

Transient laminar flow in a narrow slit

A Newtonian fluid of constant viscosity μ and constant density ρ is initially at rest between two large horizontal plates a distance $2B$ apart.



At time $t = 0$ a constant pressure gradient $\frac{\Delta p}{L} = \frac{p(z=L) - p(z=0)}{L}$ is imposed on the system, which sets the fluid in motion. Here, we study how the transient velocity profile develops with time.

- Using the relevant equation of motion, show that the velocity profile is governed by the following PDE.

$$\rho \frac{\partial v_z}{\partial t} = \mu \frac{\partial^2 v_z}{\partial x^2} - \frac{\Delta p}{L}$$

- Using no-slip boundary conditions, demonstrate that the steady-state velocity profile in the above system is as follows:

$$v_{z\infty} = v_{max} \left[1 - \left(\frac{x}{B} \right)^2 \right]$$

, where $v_{max} = \frac{B^2 \Delta p}{2\mu L}$

- Demonstrate that the partial differential equation derived in part 1 can be expressed in a dimensionless form as:

$$\frac{\partial \phi}{\partial \tau} = \frac{\partial^2 \phi}{\partial \eta^2} + 2$$

, where $\phi \equiv \frac{v_z}{v_{max}}$, $\tau = \frac{\mu t}{\rho B^2}$, and $\eta \equiv \frac{x}{B}$

- Show that the steady-state velocity profile in dimensionless form is $\phi_\infty = 1 - \eta^2$.

- 5- Write down the initial condition and two boundary conditions for $\phi = \phi(\eta, \tau)$

$$\phi(\eta, \tau = 0) = \dots$$

$$\phi(\eta = -1, \tau) = \dots$$

$$\phi(\eta = +1, \tau) = \dots$$

- 6- With the above boundary conditions, ϕ can be expanded in terms of cosine functions as:

$$\phi(\eta, \tau) = \sum_{n=1}^{\infty} C_n(\tau) \cos\left(\frac{2n-1}{2}\pi\eta\right)$$

$\sin\left(\frac{2n-1}{2}\pi\right) = (-1)^{n-1}$

Substituting the above summation in the partial differential equation $\frac{\partial\phi}{\partial\tau} = \frac{\partial^2\phi}{\partial\eta^2} + 2$, derive an ordinary differential equation for $C_n(t)$ and evaluate the integration constants using the proper orthogonality equation for the cosine functions.

- 7- Plot ϕ as a function of η for $\tau = 0.01, 0.05, 0.1, 0.5, 1.0, 2.0$, and 3.0 .

1. Governing PDE.

- For incompressible Newtonian flow in $\begin{cases} z\text{-direction} \\ z\text{-momentum} \end{cases}$

$$\rho \frac{\partial v_z}{\partial t} = -\frac{\partial p}{\partial z} + \mu \frac{\partial^2 v_z}{\partial x^2} \quad \Rightarrow \quad \rho \frac{\partial v_z}{\partial t} = \mu \frac{\partial^2 v_z}{\partial x^2} - \frac{\Delta p}{L}$$

but $\frac{\partial p}{\partial z} = \frac{\Delta p}{L}$

2. Steady state velocity profile.

$$\frac{\partial v_z}{\partial t} = 0 \text{ (s.s.)} \longrightarrow \mu \frac{d^2 v_{z,\infty}}{dx^2} = \frac{\Delta p}{L}$$

$$\Rightarrow \frac{d^2 v_{z,\infty}}{dx^2} = \frac{1}{\mu} \frac{\Delta p}{L}$$

$$\Rightarrow \frac{dv_{z,\infty}}{dx} = \int \frac{1}{\mu} \frac{\Delta p}{L} dx = \frac{\Delta p}{\mu L} x + C_1$$

$$\Rightarrow v_{z,\infty} = \int \left(\frac{\Delta p}{\mu L} x + C_1 \right) dx = \frac{\Delta p}{2\mu L} x^2 + C_1 x + C_2$$

• @ $x = 0 \rightarrow C_1 = 0$

• No-slip at the walls: $v_{z,\infty}(\pm B) = 0$

• @ $x = B \rightarrow 0 = \frac{\Delta p}{2\mu L} B^2 + C_2 \Rightarrow C_2 = -\frac{\Delta p}{2\mu L} B^2$

$$\Rightarrow v_{z,\infty}(x) = \frac{\Delta p}{2\mu L} \left(x^2 - B^2 \right) = \frac{\Delta p}{2\mu L} B^2 \left[\left(\frac{x}{B} \right)^2 - 1 \right]$$

$$\Rightarrow v_{z,\infty}(x) = -\frac{B^2 \Delta p}{2\mu L} \left[1 - \left(\frac{x}{B} \right)^2 \right]$$

$v_{\max}$

$$\Rightarrow v_{z,\infty} = v_{\max} \left[1 - \left(\frac{x}{B} \right)^2 \right]$$

$$\Rightarrow -\frac{\Delta p}{L} \frac{B^2}{\mu v_{\max}} = 2$$

3. Dimensionless form.

$$\phi = \frac{v_z}{v_{\max}} \rightarrow v_z = v_{\max} \phi \quad \Rightarrow \quad \frac{\partial v_z}{\partial t} = v_{\max} \frac{\partial \phi}{\partial t}$$

$$\tau = \frac{\mu t}{\rho B^2} \rightarrow \frac{\partial \tau}{\partial t} = \frac{\mu}{\rho B^2} \rightarrow \frac{\partial \phi}{\partial t} = \frac{\partial \phi}{\partial \tau} \frac{\mu}{\rho B^2} \quad \Rightarrow \quad \frac{\partial v_z}{\partial t} = v_{\max} \frac{\mu}{\rho B^2} \frac{\partial \phi}{\partial \tau}$$

$$\eta = \frac{x}{B} \Rightarrow \frac{\partial}{\partial x} = \frac{1}{B} \frac{\partial}{\partial \eta}$$

$$\frac{\partial^2 v_z}{\partial x^2} = \frac{v_{\max}}{B^2} \frac{\partial^2 \phi}{\partial \eta^2}$$

Substitute into PDE:

$$\Rightarrow \frac{\partial \phi}{\partial \tau} = \frac{\partial^2 \phi}{\partial \eta^2} - \frac{\Delta p B^2}{L \mu v_{\max}}$$

$$\Rightarrow \frac{\partial \phi}{\partial \tau} = \frac{\partial^2 \phi}{\partial \eta^2} + 2$$

4. Steady-state in dimensionless form.

From part 2: $1 - \left(\frac{x}{B} \right)^2 = \frac{v_{z,\infty}}{v_{\max}}$

$$\phi_{\infty} = \frac{v_{z,\infty}}{v_{\max}} = 1 - \eta^2$$

5. Initial and boundary conditions.

- Initial the fluid is at rest: $v_z(x, 0) = 0 \Rightarrow \phi(\eta, 0) = 0$
- No-slip at walls $x = \pm B$, so $v_z(\pm B, t) = 0 \Rightarrow \phi(\pm 1, \tau) = 0$

6. Solve using cosine expansion

$$\phi(\eta, \tau) = \sum_{n=1}^{\infty} C_n(\tau) \cos\left(\frac{(2n-1)\pi}{2} \eta\right)$$

$$\lambda_n = \frac{(2n-1)\pi}{2}$$

$$\Rightarrow \phi(\eta, \tau) = \sum_{n=1}^{\infty} C_n(\tau) \cos(\lambda_n \eta)$$

$$\frac{\partial \phi}{\partial \tau} = \sum_{n=1}^{\infty} C_n'(\tau) \cos(\lambda_n \eta) \quad (\text{PDE}) \quad \left\{ \begin{array}{l} \sum_{n=1}^{\infty} C_n'(\tau) \cos(\lambda_n \eta) \\ \sum_{n=1}^{\infty} -\lambda_n^2 C_n(\tau) \cos(\lambda_n \eta) + 2 \end{array} \right.$$

$$\Rightarrow \frac{\partial^2 \phi}{\partial \eta^2} = \sum_{n=1}^{\infty} -\lambda_n^2 C_n(\tau) \cos(\lambda_n \eta) = \sum_{n=1}^{\infty} -\lambda_n^2 C_n(\tau) \cos(\lambda_n \eta) + 2$$

$$\Rightarrow 2 = \sum_{n=1}^{\infty} A_n \cos(\lambda_n \eta)$$

Applying orthogonality on $[-1, 1]$: $A_n = \frac{\int_{-1}^1 2 \cos(\lambda_n \eta) d\eta}{\int_{-1}^1 \cos^2(\lambda_n \eta) d\eta} \rightarrow 2 \left[\frac{\sin(\lambda_n \eta)}{\lambda_n} \right]_{-1}^1 = \frac{4 \sin(\lambda_n)}{\lambda_n}$

$$\Rightarrow A_n = \frac{4(-1)^{n-1}}{\lambda_n} = C_n' + \lambda_n^2 C_n \quad (\text{ODE})$$

Solve ODE:

$$C_n(\tau) = K_n e^{-\lambda_n^2 \tau} + \frac{4(-1)^{n-1}}{\lambda_n^3}$$

$$\phi(\eta, 0) = 0 \rightarrow C_n(0) = \cos(0) \Rightarrow 0 = K_n + \frac{4(-1)^{n-1}}{\lambda_n^3}$$

$$\Rightarrow K_n = -\frac{4(-1)^{n-1}}{\lambda_n^3}$$

$$\Rightarrow C_n(\tau) = \frac{4(-1)^{n-1}}{\lambda_n^3} (1 - e^{-\lambda_n^2 \tau})$$

$$\lambda_n = \frac{(2n-1)\pi}{2} \Rightarrow \phi(\eta, \tau) = \sum_{n=1}^{\infty} \frac{4(-1)^{n-1}}{\lambda_n^3} (1 - e^{-\lambda_n^2 \tau}) \cos(\lambda_n \eta)$$

A more common simplified form is obtained using: $\frac{4}{\lambda_n^3} = \frac{32}{(2n-1)^3 \pi^3}$

$$\phi_0: \phi(\eta, \tau) = \sum_{n=1}^{\infty} \frac{32(-1)^{n-1}}{(2n-1)^3 \pi^3} \left[1 - \exp\left(-\frac{(2n-1)^2 \pi^2}{4} \tau\right) \right] \cos\left[\frac{(2n-1)\pi}{2} \eta\right]$$